HOW POPULATIONS EVOLVE

Module map

- Module 13: How Populations Evolve
- Connected lab: lab-13_how-populations-evolve.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 13

How Populations Evolve

1. Allele frequencies as the unit of change
2. Sources of variation
3. Five mechanisms of microevolution
4. Genetic drift and selection
5. Hardy-Weinberg as a baseline

Lab evidence

lab-13_how-populations-evolve.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

HOW POPULATIONS EVOLVE

Learning objectives

- Define microevolution using allele frequencies.
- Compare the five mechanisms that change allele frequencies.
- Distinguish bottleneck and founder effects.
- Interpret directional, stabilizing, and disruptive selection.
- Use Hardy-Weinberg as a non-evolving baseline.

OBJECTIVE 1
Define microevolution using allele frequencies.

OBJECTIVE 2
Compare the five mechanisms that change allele frequencies.

OBJECTIVE 3
Distinguish bottleneck and founder effects.

OBJECTIVE 4
Interpret directional, stabilizing, and disruptive selection.

OBJECTIVE 5
Use Hardy-Weinberg as a non-evolving baseline.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

HOW POPULATIONS EVOLVE

Topic sequence

- Allele frequencies as the unit of change: Microevolution is change in allele frequencies across generations.
- Sources of variation: Mutation creates new variation, while recombination reshuffles existing variation.
- Five mechanisms of microevolution: Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.
- Genetic drift and selection: Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.
- Hardy-Weinberg as a baseline: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

01
Allele frequencies as the unit of change
Microevolution is change in allele frequencies across generations.

02
Sources of variation
Mutation creates new variation, while recombination reshuffles existing variation.

03
Five mechanisms of microevolution
Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.

04
Genetic drift and selection
Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.

05
Hardy-Weinberg as a baseline
Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

HOW POPULATIONS EVOLVE

Module 13: How Populations Evolve Evidence Map

- Module focus: How Populations Evolve
- Central claim: How Populations Evolve explains allele-frequency change through drift, gene flow, selection, and baseline models.
- Key nodes to connect: Allele frequencies as the unit of change, Sources of variation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-13_how-populations-evolve.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

HOW POPULATIONS EVOLVE

Module 13: How Populations Evolve Reasoning Sequence

- Module focus: How Populations Evolve
- Trace the model: Allele frequencies as the unit of change -> Sources of variation -> Five mechanisms of microevolution -> Genetic drift and selection
- Inputs become outputs: Allele frequencies as the unit of change, Sources of variation -> Define microevolution using allele frequencies., Compare the five mechanisms that change allele frequencies.
- Feedback check: lab-13_how-populations-evolve.md evidence checks whether students can explain

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

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Terms as evidence handles

- Allele frequency: Proportion of a particular allele in a population.
- Microevolution: Allele-frequency change across generations.
- Mutation: DNA sequence change that can create new alleles.
- Gene flow: Movement of alleles between populations.
- Genetic drift: Random allele-frequency change, strongest in small populations.
- Bottleneck effect: Drift after a sharp population reduction.

ALLELE FREQUENCY
Proportion of a particular allele in a population.

MICROEVOLUTION
Allele-frequency change across generations.

MUTATION
DNA sequence change that can create new alleles.

GENE FLOW
Movement of alleles between populations.

GENETIC DRIFT
Random allele-frequency change, strongest in small populations.

BOTTLENECK EFFECT
Drift after a sharp population reduction.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-13_how-populations-evolve.md
- Lab evidence should test or illustrate: Five mechanisms of microevolution
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Allele frequencies as the unit of change

EVIDENCE
lab-13_how-populations-evolve.md

CLAIM
Define microevolution using allele frequencies.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: Microevolution is change in allele frequencies across generations.
- Stronger explanation: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Microevolution is change in allele frequencies across generations.

DEEPER
Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

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Module 13: How Populations Evolve Retrieval and Lab Check

- Module focus: How Populations Evolve
- Retrieval prompt: What is an allele frequency?
- Answer check: Define microevolution using allele frequencies.
- Required terms: Allele frequency, Microevolution, Mutation
- Lab connection: lab-13_how-populations-evolve.md; lab-13_how-populations-evolve.md supplies evidence for allele-frequency changes across generations.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Microevolution is defined as:
- 2. Genetic drift has its strongest effect in:
- 3. A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:
- 4. The Hardy-Weinberg principle is useful mainly because it:

QUESTION 1

Key: B

Microevolution is defined as:

QUESTION 2

Key: A

Genetic drift has its strongest effect in:

QUESTION 3

Key: D

A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:

QUESTION 4

Key: C

The Hardy-Weinberg principle is useful mainly because it:

Teaching note: Use the quiz as formative feedback. Answer key: 1=B, 2=A, 3=D, 4=C

HOW POPULATIONS EVOLVE

Synthesis and exit ticket

- Exit claim: explain allele frequencies as the unit of change using evidence from lab-13_how-populations-evolve.md.
- Must include: Allele frequency, Microevolution, Mutation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Allele frequencies as the unit of change

EVIDENCE
lab-13_how-populations-evolve.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.